MEDICAL ROBOTICS — MASTER'S PROJECT

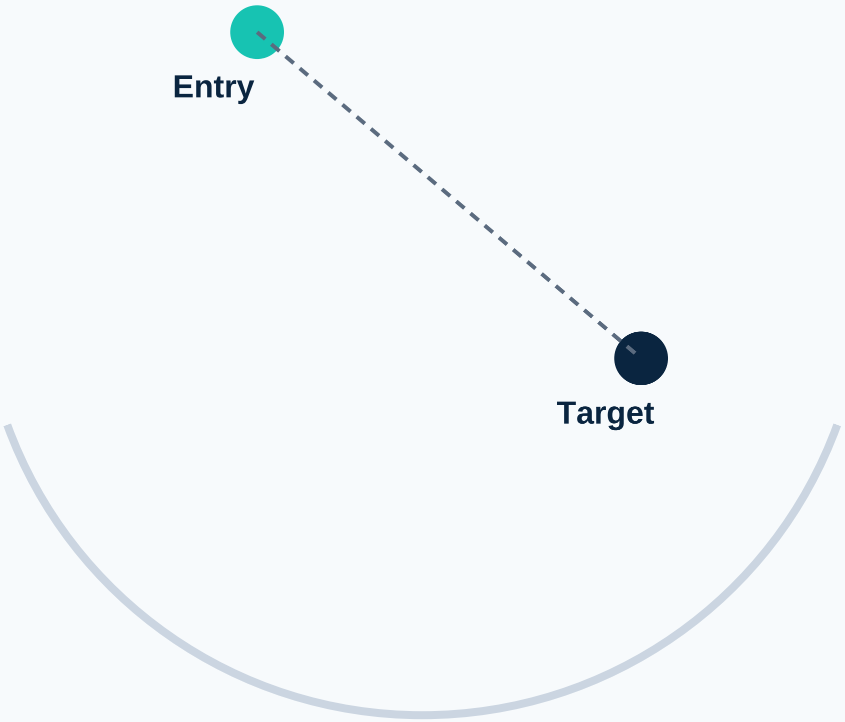
Multi-RCM Kinematic Control for a ROSA-like Surgical Robot

A Unity simulation of double Remote-Center-of-Motion control

Samuele Civale · Medical Robotics, 2026

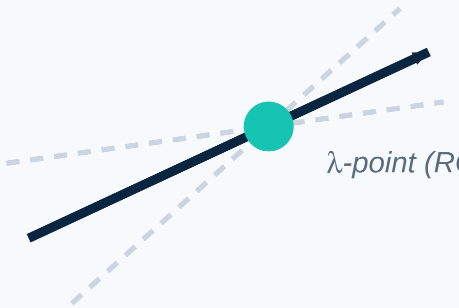
One Entry Point. Zero Lateral Force.

Stereotactic neurosurgery inserts a rigid tool through a single skull opening toward a deep target.



Any sideways force at that opening risks tissue damage.

A Pivot Point That Isn't Physically There



$$p\_RCM(q, \lambda) = p\_base(q) + \lambda \cdot L \cdot z(q)$$

λ slides the RCM point along the tool shaft as insertion depth changes.

03 — WHY IT MATTERS

Constraining the Pivot Changes Everything

X

Without RCM

Shaft levers against the entry point as the arm reconfigures → lateral tissue load at the skull surface.

✓

With RCM

Tip reaches the target while the shaft pivots cleanly through the entry point, simultaneously.

Project at a Glance

6	4	1	9 / 9
DOF procedural arm	Task modes	Controller, built from scratch	Validation checks passed

Four Tasks, One Controller

Task 1

Entry-RCM + Tip Target

Task 2

Target-RCM + Entry-Side Cone

Task 3

Insertion Sequence (main task)

Task 4

Entry-RCM + Tip Cone Around Target

Inside Task 3: The Insertion Sequence

ApproachEntry

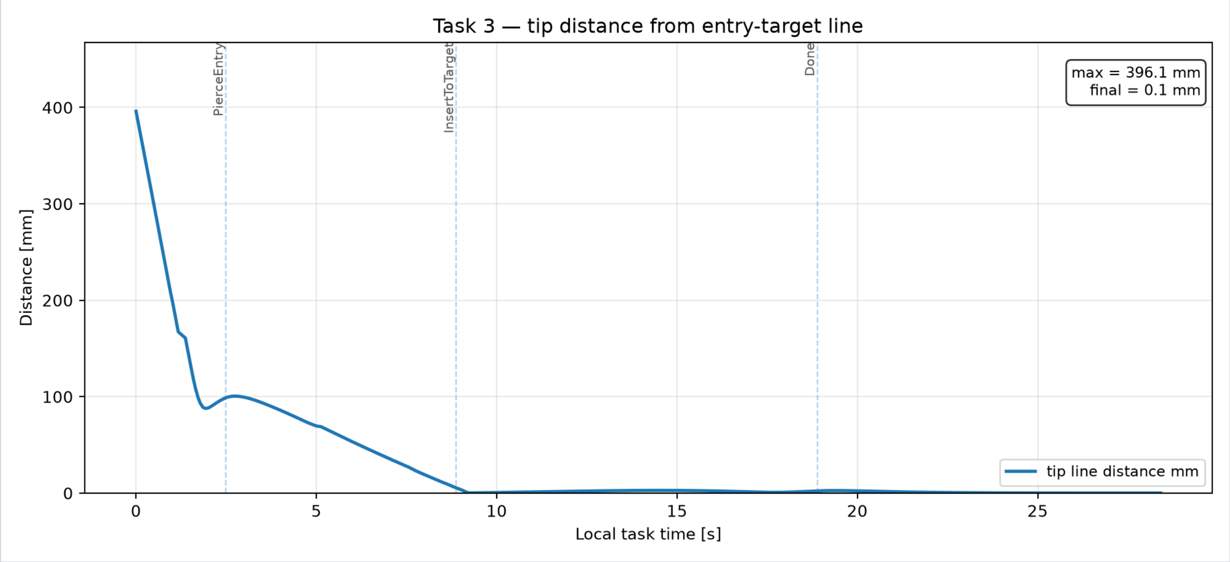
Align needle axis, approach outside the trocar

PierceEntry

Tip travels to the entry point (RCM not yet active)

InsertToTarget

RCM locked at entry — tip advances to target



Tip-to-corridor distance collapses to ~0 mm once insertion begins.

Controller Architecture

Stacked Tasks

Tip / cone / RCM tasks stacked into one Jacobian

→

Damped Least Squares

Custom Gauss–Jordan solve

→

Null-space Correction

Joint limits, λ target, secondary objectives

→

Velocity Output

Saturated joint + λdot velocities, integrated

Jacobians computed by numerical finite differencing.

08 — CONTROLLER (MATH)

The Control Law, in Full

Stacked task Jacobian

```
[ xdot_task ]   [ J_task   0 ] [ qdot  ]
[ xdot_RCM  ] = [ J_RCM      ] [ λdot ]
```

Per-task error → velocity

```
ydot_i = K_i · (x_des,i − x_i),  saturated
```

Damped least-squares solve

```
[ qdot ; λdot ] = J^T (J J^T + μ²I)^-1 y
```

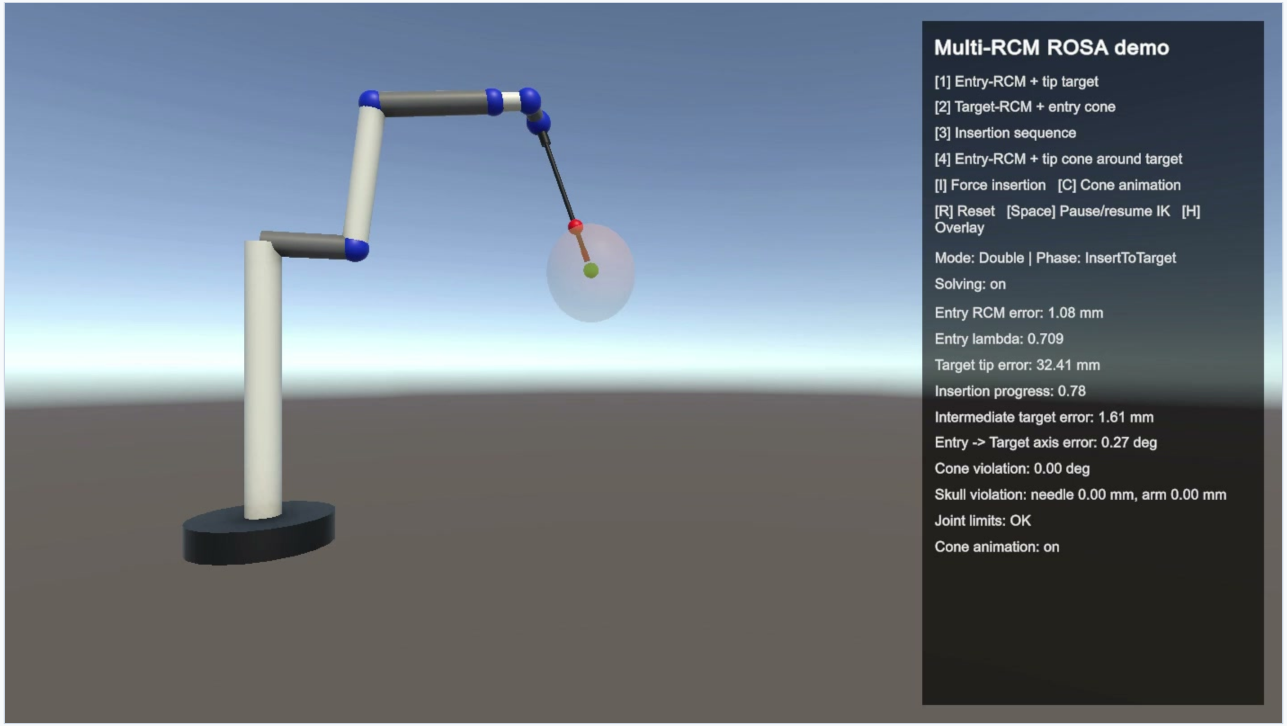
Null-space secondary objective

```
+ (I − J+ J) · w
w: joint limits, λ target
```

Integration (saturated): q_k+1 = clamp(q_k + qdot·dt) · λ_k+1 = clamp(λ_k + λdot·dt, 0.02, 0.995)

Built in Unity — Three Scripts

- Project4SceneBuilder.cs — builds the scene procedurally
- ROSADoubleRCMController.cs — kinematics, control law, logging
- FreeFlyCameraKeyboard.cs — keyboard fly-camera



Simulation view: procedural arm, needle, target marker, live HUD. (Captured from an earlier development build.)

From Simulation to Evidence

Unity Controller

Live CSV log
~20 Hz, every relevant variable

→

Python Analyzer

analyze_rcm_logs_clean.py
one metric per plot

→

Plots + Tables

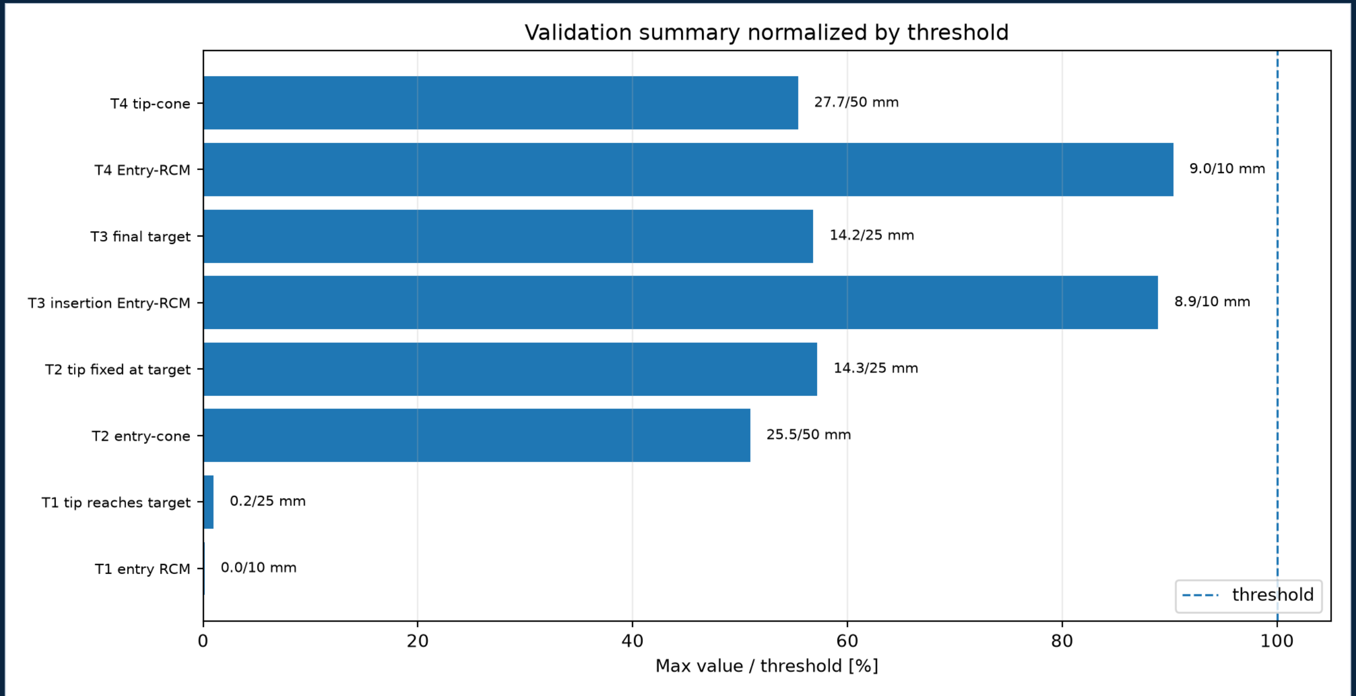
Thresholded plots +
pass_fail_checks.csv

Results: Everything Passes

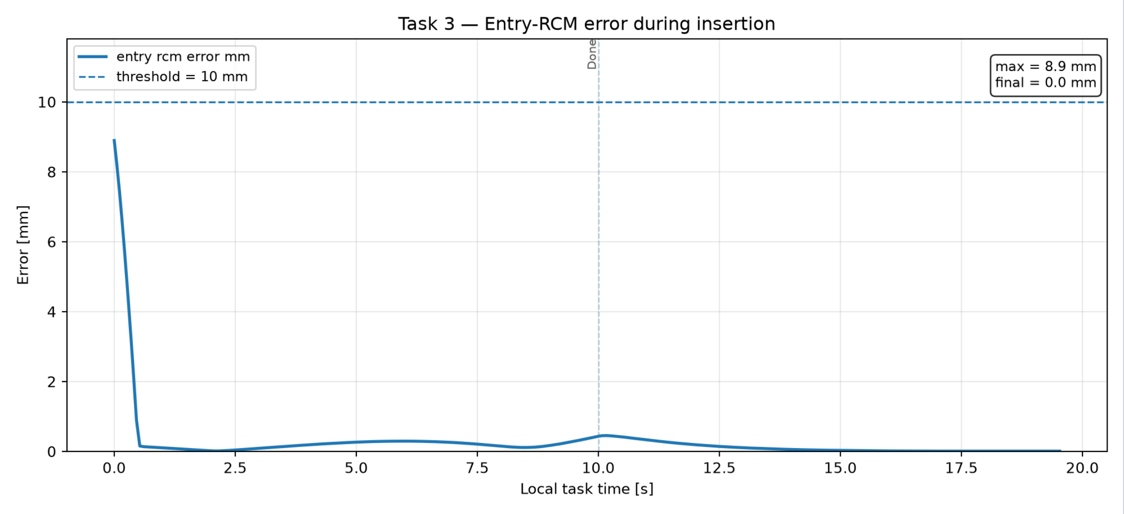
9 / 9

validation checks passed

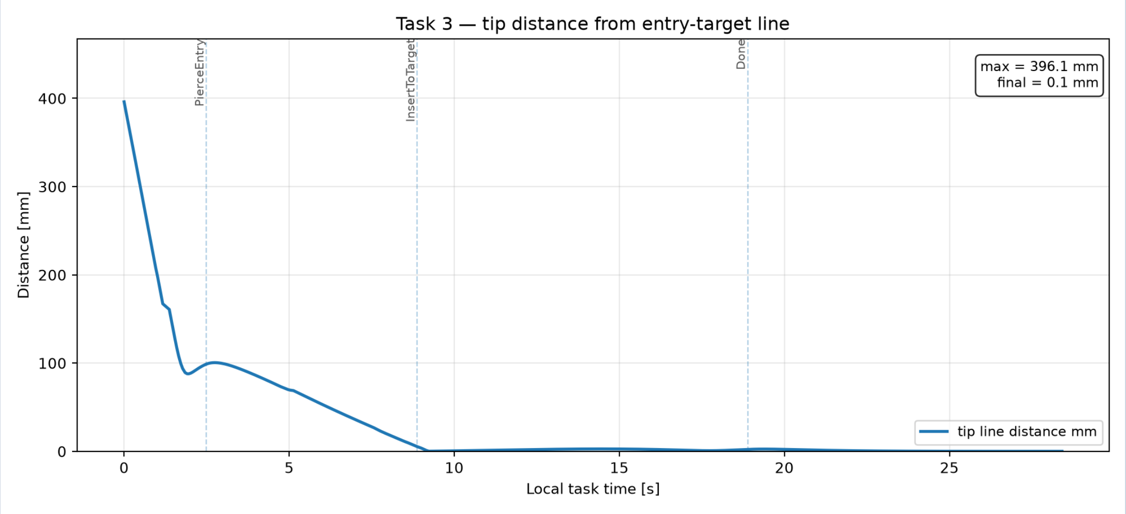
Tightest margin: entry-RCM error during insertion, ~90% of threshold.



Results: Safe Insertion (Task 3)

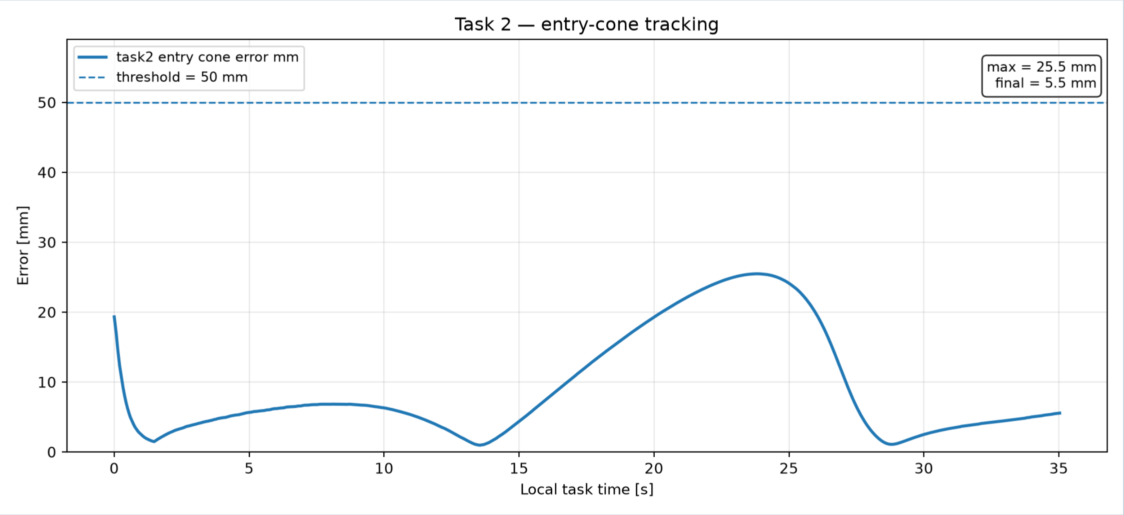


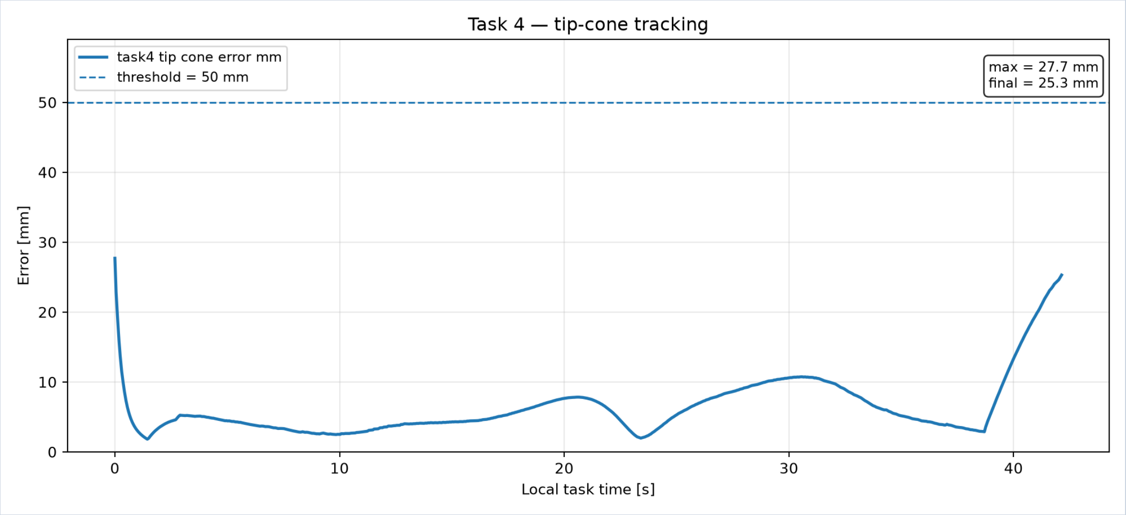
Entry-RCM error → ~0 mm once insertion starts



Tip stays on the planned entry–target line

Tracking Moving Targets (Tasks 2 & 4)





Bounded, oscillating tracking error — expected for a continuously rotating target, both well under threshold.

Limitations

- ROSA-like, not an official CAD model — plausible DH, not measured
- Needle = rigid cylinder, no tissue interaction
- No anatomical collision avoidance in the final build
- Purely kinematic — no dynamics, no force control
- Jacobians computed numerically, not symbolically

What's Next

1. Analytic Jacobian (replace finite differences)
2. Re-introduce anatomy as a separate validation layer
3. Real ROSA DH parameters, if available
4. Randomized multi-trial validation

CONCLUSION

A Validated Double-RCM Controller, Honestly Scoped

Four tasks. One controller built from scratch. Nine of nine checks passed.
Anatomy deliberately left out to keep the focus on the RCM formulation.

Thank you — questions welcome.